



For the questions a, b, c, d & e consider the circuit shown in figure 1 and answer (use nodal method):

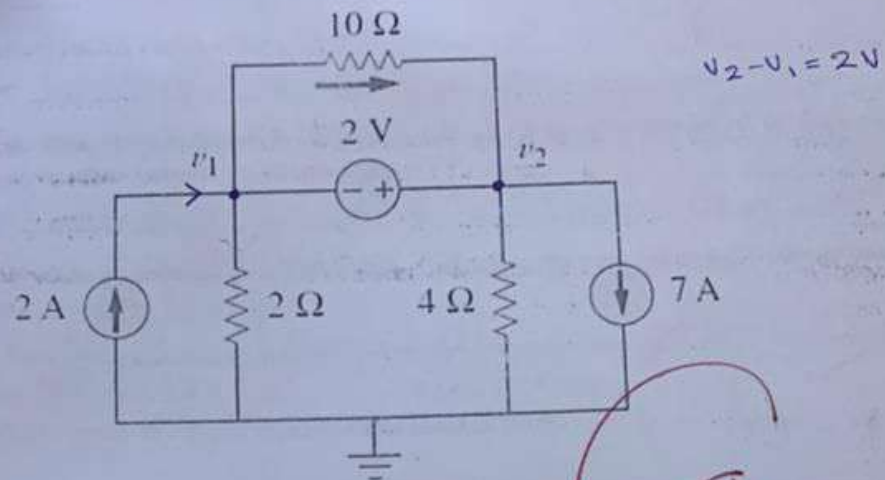


Fig.1

a. The value of  $V_1$  is:

$$V_1 = -7.33 \text{ V}$$

b. The value of  $V_2$  is:

$$V_2 = -5.33 \text{ V}$$

c. The value of the current (i) through  $10\Omega$  resistor is:

$$i = 0.2 \text{ A}$$

c. The power absorbed by  $4\Omega$  resistor is:

$$P = \frac{V^2}{R} = \frac{(V_2)^2}{4} = \frac{(-5.33)^2}{4} = 7.11 \text{ W}$$

d. The power absorbed by  $2 \text{ A}$  current source is:

$$P = I \cdot V_1 = 2 \cdot (-5.33) = -10.66 \text{ W}$$

e. Write the required equations (do not simplify)

$$V_2 - V_1 = 2 \quad \text{--- (1)}$$

$$2 = \frac{7}{2} + \frac{V_1}{2} + \frac{V_2}{4} \quad \text{--- (2)}$$

Prob. 2: Consider the circuit shown in figure 2. and answer:

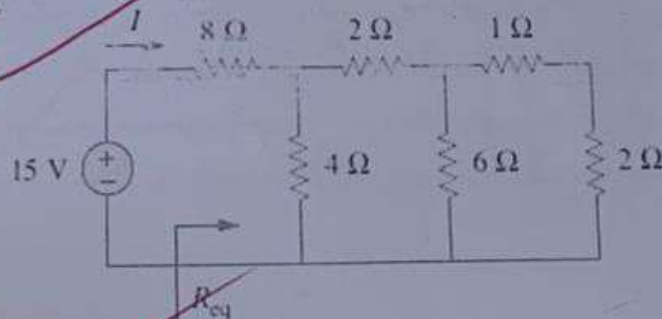


Fig.2

a. The equivalent resistance  $R_{eq}$  is:

$$R_{eq} = 10 \Omega$$



b. The current  $I$  is:

$$I = \frac{V}{R_{eq}} = \frac{15}{10} = 1.5A$$

c. The Power absorbed by  $4\Omega$  resistor is:

$$P = I^2 R = (0.75)^2 (4) = 2.25W$$

Prob.3: a. The charge transferred through an element is given by  $q(t) = 2t^3 + 5t^2 + 6$  C. Calculate the current entering the element at instant  $t = 10$  ms.

$$i = \frac{dq}{dt} \Rightarrow i = 6t^2 + 10t \Rightarrow i|_{t=10ms} = 6(10 \times 10^{-3})^2 + 10(10) \times 10^{-3} = 600 \times 10^{-6} + 100 \times 10^{-3} = 100.6mA$$

b. A load draws 500W when connected to a 230 V line. Calculate the current and the energy consumed in two hours in joules?

$$P = 500W, V = 230V, t = 2 \text{ hours}$$

$$i = \frac{P}{V} = \frac{500}{230} = 2.17A$$

$$Energy = Pt = 500 \times 2 \times 60 \times 60 = 360000 = 3.6 \times 10^5 J$$

Prob.4: Refer to figure 3; write the mesh equations in matrix form:

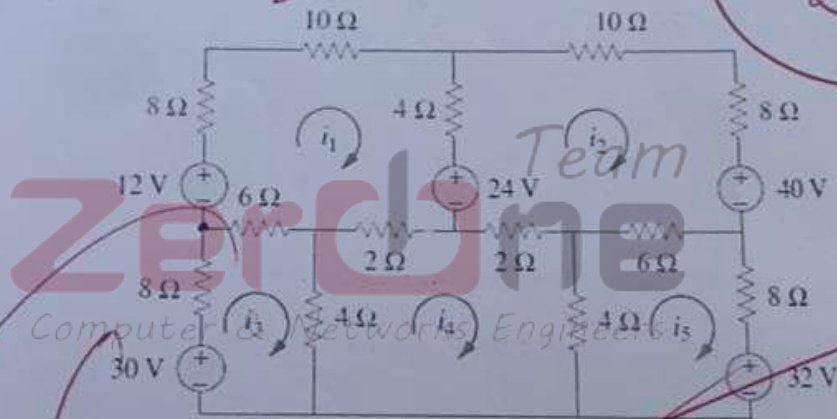


Fig. 3

$$\begin{bmatrix} 30 & -4 & -6 & -2 & 0 \\ -4 & 30 & 0 & -2 & -6 \\ -6 & 0 & 18 & -4 & 0 \\ -2 & -2 & -4 & 8 & 0 \\ 0 & -6 & 0 & -4 & 18 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} -12 \\ -16 \\ 30 \\ 0 \\ 32 \end{bmatrix}$$

Good luck